

From Vibe Coding to Spec-Driven Agentic Engineering

A Hands-On Implementation using Claude Code

PART I

Instructor: Muhammad Arif Butt, PhD

About The Instructor



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- **Education:**
  - Graduation from Pakistan Military Academy, Kakul
  - MPhil CS from University of Punjab, Lahore
  - PhD CS from University of Punjab, Lahore
- **Experience:**
  - Served in field/staff/instructional posts in Pakistan Army
  - Assistant Professor, Department of Data Science
  - Founder Excaliat (https://excaliat.com/en)
  - Founder FalconHunt (https://falconhunt.org/)
  - Co-Founder Tbox Solutionz (https://tboxsolutionz.com/)
- **Research Interest:**
  - Embedded and Real-Time Operating Systems
  - Vulnerability Analysis, Binary Exploitation & Exploit Development
  - AI-Driven Cybersecurity and Securing AI systems

Scheme of Sessions

Part- I: Overview of Agentic Coding Assistants

Part– II: Claude Ecosystem, Installing & using Claude Code

Part– III: Context Window, Slash Commands & Claude Directory

Part– IV: A Vulnerable Web Application & its Attack Vectors

Part– V: Workflow of Implementing Features & Fixing Bugs

Part– VI: Skill Market place and Writing your Own Skills

Part– VII: Creating and using Subagents and Agent Teams

Part– VIII: Model Context Protocol

Part– IX: Creating and using Hooks

Part– X: Creating and using Plugins

Part 1
Overview of Agentic Coding Assistants

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The Shift in Software Development



Traditional Development

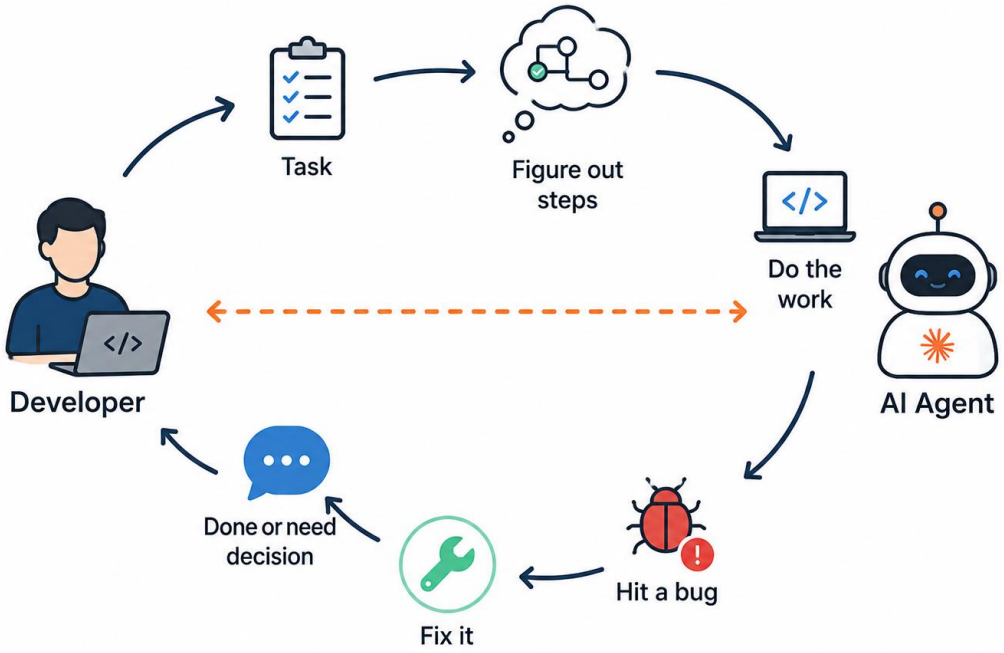
- Humans write everything manually
- IDEs provide autocomplete only
- Context switching is expensive
- Boilerplate consumes time

AI-Assisted Development

- Natural language becomes part of development
- AI can generate, refactor, debug, document, and test
- Developers move from “code writers” to “system orchestrators”

What is Vibe Coding?





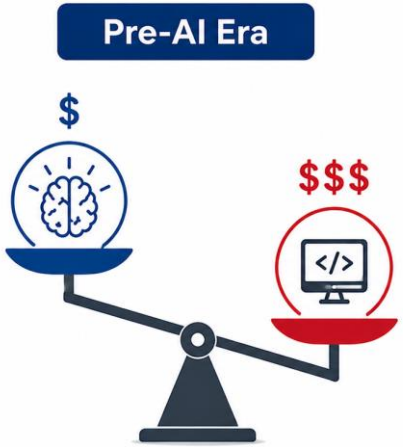
Limitations of vibe coding

- Fails in multi-team development
- Architectural in-consistency
- Poor scalability
- Results in in-secure and vulnerable code

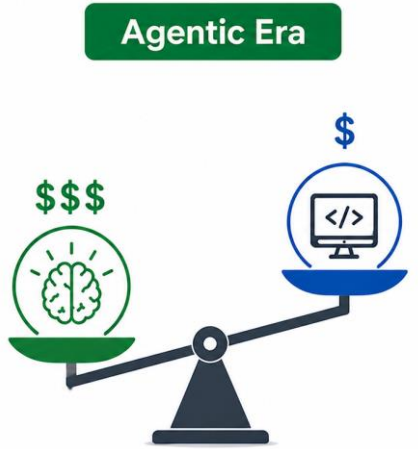
Vibe coding optimizes for speed of code generation, while the industry optimizes for collaboration, reliability, maintainability and scability

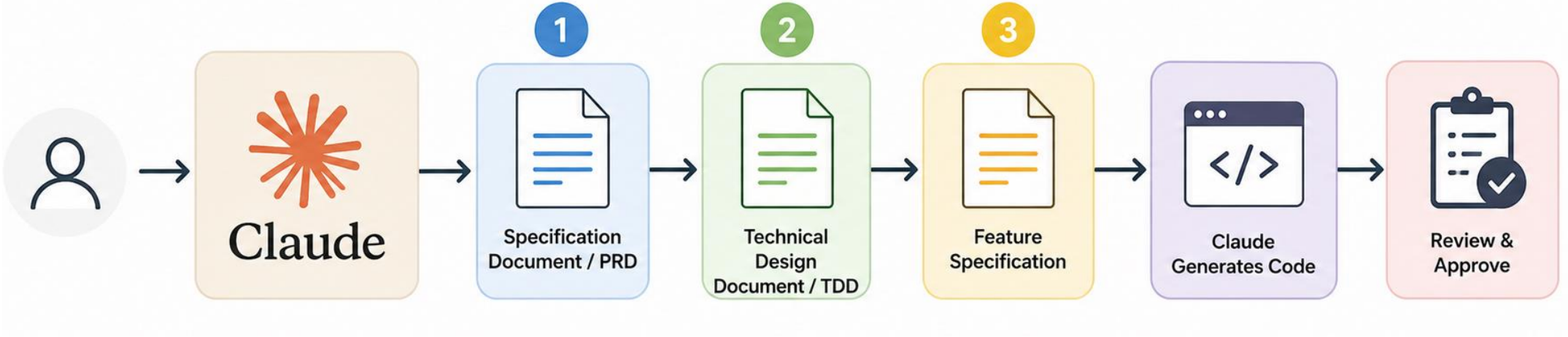
What is Spec-Driven Development (SDD)?

In the pre-AI era, thinking was cheap and coding was expensive



In the agentic era, coding is cheap and thinking is expensive





Specification-Driven Development with Claude Code

Authentication Example — 1 PRD → 1 TDD → N Feature Specs

STEP 1

□ PRD (1 document)

Product Requirements Document

"A user must be authenticated before accessing the service."

- Business requirement (the WHY)
- User-facing outcome
- No tech details, no implementation

STEP 2

□ TDD (1 document)

Technical Design Document

- Signup mechanism (email + bcrypt hashing)
- Login flow (JWT, refresh tokens)
- Session storage (Redis, HTTP-only cookies)
- Password reset (token + email service)
- Security: rate limiting, CSRF, OWASP controls
- Data model, API contracts, error handling

STEP 3

□□□ Feature Specs (N docs)

One per feature — 5 in this example

- Self-contained, implementable units
- Acceptance criteria + edge cases
- Reference PRD & TDD for context
- Each ready for Claude to implement

→ *See breakdown below*

The 5 Feature Specifications generated from the auth PRD + TDD

Feature 1: Signup Flow

Form & Validation
- Email, password, confirm password
- Strength rules, format validation

Success Path
- Create account → redirect to onboarding

Edge Cases
- Duplicate email, weak password
- Network failure, server error

Acceptance Criteria
- Confirmation email sent within 5s

Feature 2: Login Flow

Credential Entry
- Email + password fields
- "Remember me" checkbox

Success Path
- Issue JWT + refresh token cookie

Edge Cases
- Wrong password, account locked
- Rate limit after 5 failed attempts

Acceptance Criteria
- Login completes < 2s under load

Feature 3: Password Reset

Reset Request
- Email entry → send reset link
- Token expires in 15 minutes

Success Path
- User sets new password → auto-login

Edge Cases
- Expired token, unknown email
- Token reuse attempts blocked

Acceptance Criteria
- All old sessions invalidated

Feature 4: Session Mgmt

Token Lifecycle
- Access token: 15 min TTL
- Refresh token: 7 day TTL

Success Path
- Silent refresh; user stays logged in

Edge Cases
- Concurrent sessions, device limits
- Stolen token revocation

Acceptance Criteria
- Refresh handled transparently

Feature 5: Logout

Logout Action
- Single-click logout from any page
- "Logout from all devices" option

Success Path
- Clear cookies, invalidate tokens

Edge Cases
- Already expired session
- Concurrent logout requests

Acceptance Criteria
- Redirect to login page on success

STEP 4

□□ Claude Generates Code

Implements one Feature Spec at a time, grounded by PRD + TDD context

□ Review & Approve

Verify against the Feature Spec's acceptance criteria. Iterate or merge.

□ KEY INSIGHT

The number of Feature Specs is NOT fixed — it depends on how the TDD decomposes the system. For auth, 5 features is typical; for a small MVP, it might be 2–3; for an enterprise SSO system, 10+.
Rule of thumb: one Feature Spec = one self-contained, shippable unit of work that Claude can implement in a single focused session.

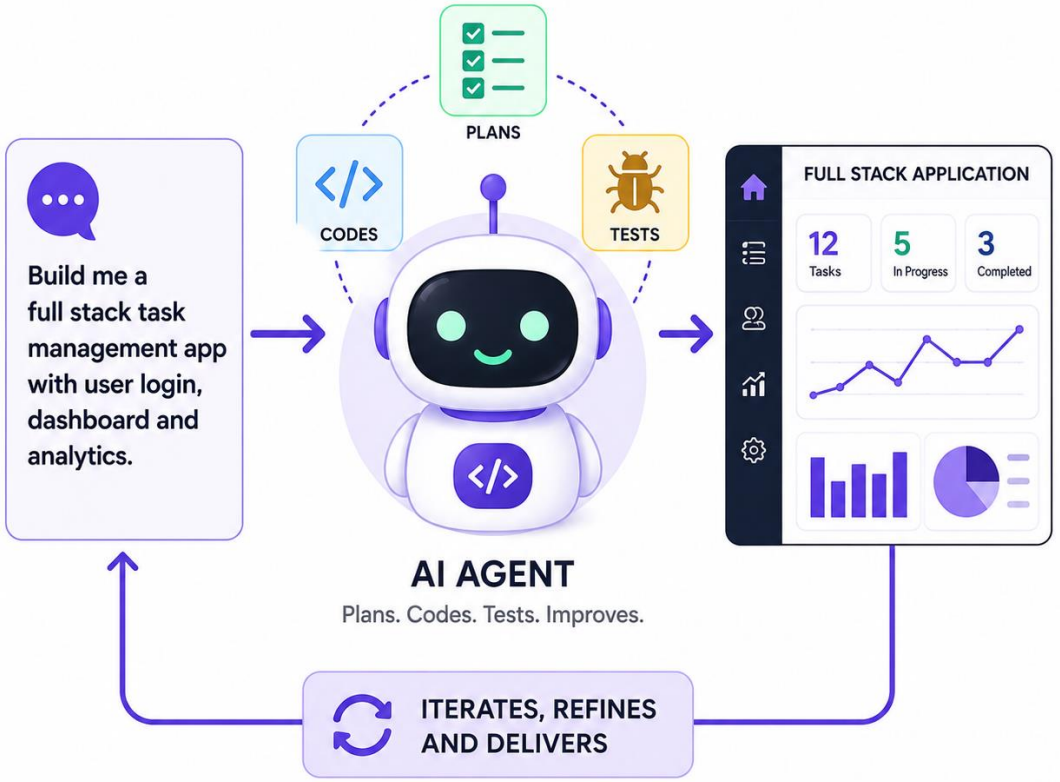
The Evolution Toward Agentic Coding

Agentic coding is an AI-driven development approach where autonomous agents can plan, reason, execute tasks, and iterate toward a defined goal while following structured workflows, engineering constraints, and system objectives.

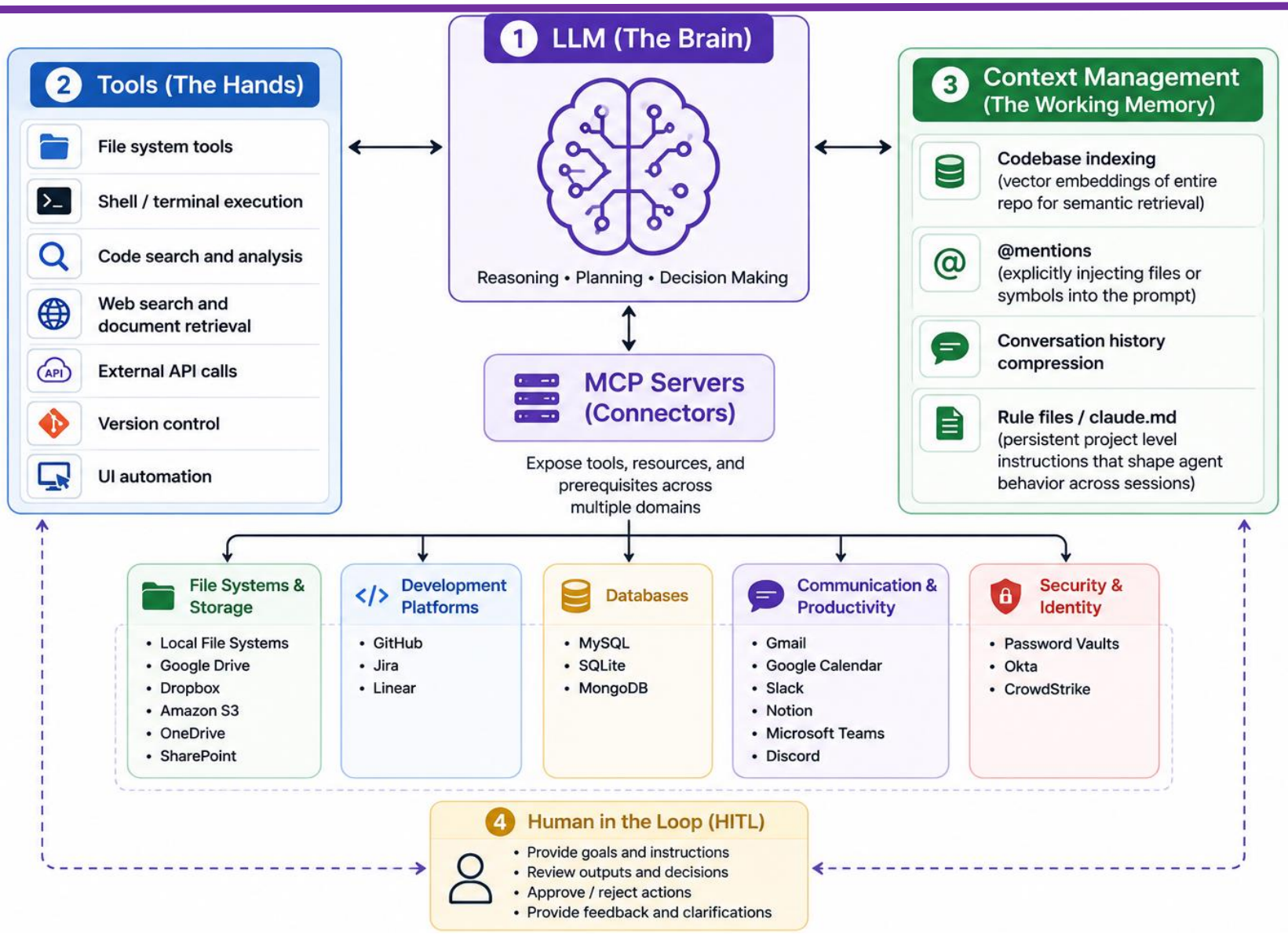
- Automates multi-step software engineering workflows.
- Improves productivity through task automation.
- Integrates with tools, APIs, and runtime environments.
- Enables contextual reasoning and intelligent decision-making.
- Advances software engineering beyond code generation.







Core Components of Agentic AI System



AI-Powered Agentic Coding Platforms



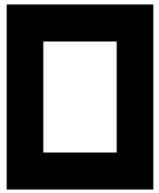
Anthropic Claude Code



OpenClaw/ ClawCode



OpenAI Codex



OpenCode



Pi



Hermes



GitHub Copilot



Windsurf



Anysphere Cursor

Q/A



Thank you!